Math 50 Midterm 2: How to Derive Every Formula on the Sheet

A companion to `midterm2_formulas.pdf` and `midterm2_notes.pdf`

How to use this document

Every formula on the formula sheet can be derived from the model equation with a small number of *moves*. Section 1 lists those moves (M1–M11) and translates each arrow on the formula sheet into a move. Every later derivation starts with a purple **Method** box saying the goal, what to start from, and which moves to use. If you can reproduce the Method box from memory, you can usually reproduce the algebra. The full steps follow, with the reason for each step in gray on the right, as in the notes.

Blue boxes hold the final result. Sections marked *(extra)* are not on the sheet itself but come from the notes and use the same methods. The last page is a cover-and-recall table: hide the right-hand columns and try to say the method for each formula.

Everything here uses only tools from the Unit 2–4 notes and the lectures (linearity of $\mathbb{E}$, bilinearity of cov, variance rules, the CLT, and sums). Nothing new is introduced.

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1 The moves, and how to read the formula sheet's arrows

1.1 The eleven moves

	Move	What you do / the rule it uses
M1	Substitute the model	Replace Y by $\beta_0 + \beta_1 X_1 + \epsilon$ (or the two-predictor version). Almost every population derivation starts here.
M2	Take $\mathbb{E}[\cdot]$	Linearity: $\mathbb{E}[aX + bY + c] = a\mathbb{E}X + b\mathbb{E}Y + c$. Use $\mathbb{E}[\epsilon] = 0$.
M3	Take $\text{cov}(X_1, \cdot)$	Bilinearity: $\text{cov}(X_1, aY + bZ + c) = a\,\text{cov}(X_1, Y) + b\,\text{cov}(X_1, Z)$. Constants drop out. $\text{cov}(X_1, X_1) = \text{var}\, X_1$. $\text{cov}(X_1, \epsilon) = 0$ because the noise is independent of the predictors.
M4	Take $\text{var}(\cdot)$	$\text{var}(aX + b) = a^2\,\text{var}\, X$. Independent pieces: variances add. In general $\text{var}(X + Y) = \text{var}\, X + \text{var}\, Y + 2\,\text{cov}(X, Y)$.
M5	Condition	"Given X" means X is a known constant. $\text{var}(Y \mid X)$ is the variance of whatever is *still random*.
M6	Rescale / standardize	Divide by standard deviations: $Z = (X - \mu)/\sigma$. Changes units, not shape.
M7	Average of N iid pieces	$\mathbb{E}[\overline{Y}] = \mu$, $\text{var}(\overline{Y}) = \sigma^2/N$. A sum of independent Normals is Normal; otherwise use the CLT for large N.
M8	Plug in	Replace each expectation by a sample average, $\mathbb{E}[\cdot] \to \frac{1}{N}\sum_i$, and $\mathbb{E}X, \mathbb{E}Y$ by $\overline{X}, \overline{Y}$.
M9	Normal → probability	Standardize, then use $\mathbb{P}(\lvert Z\rvert < 1.96) = 0.95$ and $\mathbb{P}(Z < z) = \Phi(z)$.
M10	Add and subtract	Insert $\pm\mu$ or $\pm\mathbb{E}[\hat{\theta}]$ inside a square, expand $(a + b)^2$, and show the cross term is zero.
M11	The sum trick	$\sum_i (X_i - \overline{X}) = 0$, so $\sum_i c\,(X_i - \overline{X}) = 0$ for any constant c (including $c = \overline{Y}$).

1.2 Each arrow on the sheet is a move

The labels on the arrows of the formula sheet are instructions. Here is what each one means and where it is done below.

Arrow label	From → to	What you actually do	Section
take $\text{cov}(X_1, \cdot)$ and $\mathbb{E}[\cdot]$	one predictor → coefficients	M1 M3 then divide by $\text{var}\, X_1$; M1 M2 then solve for β_0	2.1, 2.2
take $\text{var}(\cdot)$	one predictor → variance	M1 M4, noise independent of X_1	2.4
rescale by σ_{X_1}/σ_Y	coefficients → correlation	M6: multiply β_1 by σ_{X_1}/σ_Y	2.6

Arrow label	From → to	What you actually do	Section
$\text{var}(Y \mid X_1) = \sigma_\epsilon^2$	variance → variance explained	M5: given X_1, only ϵ is random	2.5
square	correlation → variance explained	square ρ, use the variance equation	2.7
omit X_2	two predictors → one predictor	the one-predictor slope of the *two*-predictor world is β_1'	3.2
$\text{var}(Y \mid X_1, X_2) = \sigma_\epsilon^2$	two predictors → variance explained	M5 with both predictors known	3.1
$\beta_1 \to \beta_1'$; β_1, β_2	coefficients / two predictors → OVB	M1 M3 in the two-predictor model	3.2
estimate β_1 from N observations	one predictor → sample dist. of $\hat{\beta}_1$	M8 to get $\hat{\beta}_1$, then M11 M2 M4	6
$\hat{\theta} = \overline{Y}$	sample distribution → sample mean	M7	4.2
$\hat{\theta} = \hat{\beta}_1$	sample distribution → sample dist. of $\hat{\beta}_1$	same recipe as $\overline{Y}$, with weights	6.4
(to bias/variance)	sample distribution → MSE	M10	4.3
± 1.96 se	sample distribution → 95% CI	M9, then rearrange the inequality	5.1
null $\beta_1 = 0$	sample dist. of $\hat{\beta}_1$ → hypothesis test	set $\beta_1 = 0$, M6 M9	5.2
(CI → test)	95% CI → hypothesis test	chain of "if and only if" steps	5.3

The general recipe for any sample distribution

Whenever you are asked "what is the sample distribution of $\hat{\theta}$?":

1. Write $\hat{\theta}$ as a (weighted) sum of the data, e.g. $\overline{Y} = \sum_i \frac{1}{N} Y_i$ or $\hat{\beta}_1 = \sum_i w_i Y_i$.
2. **Center:** take $\mathbb{E}[\cdot]$ M2. Compare with θ to get the bias.
3. **Spread:** take $\text{var}(\cdot)$ M4; independent data points, so variances add. The square root is $\text{se}(\hat{\theta})$.
4. **Shape:** a sum of independent Normals is Normal; otherwise the CLT says approximately Normal for large N M7.

If you blank on the exam

Write the model equation, then ask which arrow you need: for a **coefficient**, take $\text{cov}(X, \cdot)$; for an **intercept** or mean, take $\mathbb{E}$; for a **spread**, take var (squared coefficients, independent pieces add); for anything "**given** X", freeze X as a constant; for anything **estimated**, write it as a weighted sum of data and do center, spread, shape.

2 One predictor: the population formulas

Throughout this section the model is

$$Y = \beta_0 + \beta_1 X_1 + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2), \qquad \epsilon \text{ independent of } X_1.$$

Write $\sigma_{X_1}^2 = \text{var}(X_1)$ and $\sigma_Y^2 = \text{var}(Y)$. These are statements about the true (population) quantities, not data.

2.1 The slope: $\beta_1 = \text{cov}(X_1, Y)/\sigma_{X_1}^2$

Method

Goal: an expression for β_1 using only X_1 and Y.
Start from: the model equation.
Moves: M1 substitute the model inside $\text{cov}(X_1, Y)$; M3 split it with bilinearity; solve for β_1.

$$\begin{aligned} \text{cov}(X_1, Y) &= \text{cov}(X_1,\ \beta_0 + \beta_1 X_1 + \epsilon) && \text{(substitute the model)} \\ &= \beta_1 \,\text{cov}(X_1, X_1) + \text{cov}(X_1, \epsilon) && \text{(bilinearity; the constant } \beta_0 \text{ drops out)} \\ &= \beta_1 \sigma_{X_1}^2 + 0 && \text{(cov}(X_1, X_1) = \text{var}\, X_1\text{; } \epsilon \text{ independent of } X_1\text{)} \end{aligned}$$

Divide both sides by $\sigma_{X_1}^2$:

$$\beta_1 = \frac{\text{cov}(X_1, Y)}{\sigma_{X_1}^2}.$$

Why this move works. Taking $\text{cov}(X_1, \cdot)$ of the model "kills" everything not connected to X_1: constants vanish and the noise is uncorrelated with X_1. Only the $\beta_1 X_1$ term survives. You will use this same move for omitted-variable bias (Section 3.2) and the two-predictor system (Section 3.4).

2.2 The intercept: $\beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X_1]$

Method

Goal: β_0.
Start from: the model equation.
Moves: M1 M2 take the expectation of both sides; use $\mathbb{E}[\epsilon] = 0$; solve for β_0.

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}[\beta_0 + \beta_1 X_1 + \epsilon] && \text{(take } \mathbb{E} \text{ of both sides)} \\ &= \beta_0 + \beta_1 \mathbb{E}[X_1] + \mathbb{E}[\epsilon] && \text{(linearity)} \\ &= \beta_0 + \beta_1 \mathbb{E}[X_1] && (\mathbb{E}[\epsilon] = 0) \end{aligned}$$

$$\beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X_1].$$

Meaning. The line passes through the point of means $(\mathbb{E}X_1, \mathbb{E}Y)$. The least squares version is $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}$.

2.3 (extra) A binary predictor: β_1 is a difference of group means

Method

Goal: read β_1 off group averages when $X_1 \in \{0, 1\}$.
Start from: the model, conditioned on each value of X_1.
Moves: M5 condition on $X_1 = 1$ and on $X_1 = 0$; M2 take $\mathbb{E}$; subtract.

Given $X_1 = x$, the terms $\beta_0 + \beta_1 x$ are constants, so $\mathbb{E}[Y \mid X_1 = x] = \beta_0 + \beta_1 x + \mathbb{E}[\epsilon] = \beta_0 + \beta_1 x$. Therefore

$$\mathbb{E}[Y \mid X_1 = 1] - \mathbb{E}[Y \mid X_1 = 0] = (\beta_0 + \beta_1) - \beta_0 = \beta_1.$$

This is how the notes read slopes off plots (Lecture 7) and is the basis of the graders example and of the second OVB method in Section 3.2.

2.4 The variance equation: $\sigma_Y^2 = \beta_1^2 \sigma_{X_1}^2 + \sigma_\epsilon^2$

Method

Goal: split the variance of Y into the part from X_1 and the part from noise.
Start from: the model equation.
Moves: M1 M4 take var of both sides; the two random pieces are independent, so their variances add.

$$\begin{aligned} \sigma_Y^2 = \operatorname{var}(Y) &= \operatorname{var}(\beta_0 + \beta_1 X_1 + \epsilon) && \text{(substitute the model)} \\ &= \operatorname{var}(\beta_1 X_1) + \operatorname{var}(\epsilon) && \text{(constant } \beta_0 \text{ drops; } X_1, \epsilon \text{ independent)} \\ &= \beta_1^2 \sigma_{X_1}^2 + \sigma_\epsilon^2 && (\operatorname{var}(aX) = a^2 \operatorname{var} X) \end{aligned}$$

$$\sigma_Y^2 = \beta_1^2 \sigma_{X_1}^2 + \sigma_\epsilon^2.$$

Common slip: $\operatorname{var}(\beta_1 X_1)$ is $\beta_1^2 \sigma_{X_1}^2$, not $\beta_1 \sigma_{X_1}^2$. The coefficient comes out *squared*.

2.5 The conditional variance: $\operatorname{var}(Y \mid X_1) = \sigma_\epsilon^2$

Method

Goal: how much Y varies once X_1 is known.
Start from: the model equation.
Moves: M5 once X_1 is known, $\beta_0 + \beta_1 X_1$ is a constant; M4 a constant does not change variance.

$$\begin{aligned} \operatorname{var}(Y \mid X_1) &= \operatorname{var}(\beta_0 + \beta_1 X_1 + \epsilon \mid X_1) && \text{(substitute the model)} \\ &= \operatorname{var}(\epsilon \mid X_1) && (\beta_0 + \beta_1 X_1 \text{ is a known constant}) \\ &= \sigma_\epsilon^2 && (\epsilon \text{ independent of } X_1) \end{aligned}$$

Meaning. σ_ϵ^2 is the spread of the points around the line; σ_Y^2 is the total spread. Comparing the two is exactly what ρ^2 does.

2.6 Correlation: $\rho = \beta_1 \, \sigma_{X_1}/\sigma_Y$

Method

Goal: the slope in standard-deviation units.
Start from: either the definition $\rho = \operatorname{cov}(X_1, Y)/(\sigma_{X_1}\sigma_Y)$ (method A) or the model in standardized variables (method B).
Moves: A: substitute $\operatorname{cov}(X_1, Y) = \beta_1 \sigma_{X_1}^2$ from Section 2.1 and cancel. B: M1 M6 standardize both sides and read off the new slope.

Method A (fast, use on the exam).

$$\rho = \frac{\operatorname{cov}(X_1, Y)}{\sigma_{X_1}\sigma_Y} = \frac{\beta_1 \sigma_{X_1}^2}{\sigma_{X_1}\sigma_Y} = \beta_1 \frac{\sigma_{X_1}}{\sigma_Y}.$$

Method B (shows *why* it is called "rescaling"). Let $Z_X = (X_1 - \mu_X)/\sigma_{X_1}$ and $Z_Y = (Y - \mu_Y)/\sigma_Y$, with $\mu_Y = \beta_0 + \beta_1 \mu_X$ from Section 2.2.

$$\begin{aligned}
Z_Y = \frac{Y - \mu_Y}{\sigma_Y} &= \frac{(\beta_0 + \beta_1 X_1 + \epsilon) - (\beta_0 + \beta_1 \mu_X)}{\sigma_Y} && \text{(substitute } Y \text{ and } \mu_Y\text{)} \\
&= \frac{\beta_1 (X_1 - \mu_X) + \epsilon}{\sigma_Y} && (\beta_0 \text{ cancels}) \\
&= \frac{\beta_1 \sigma_{X_1}}{\sigma_Y} \cdot \frac{X_1 - \mu_X}{\sigma_{X_1}} + \frac{\epsilon}{\sigma_Y} && \text{(multiply and divide by } \sigma_{X_1}\text{)} \\
&= \underbrace{\frac{\beta_1 \sigma_{X_1}}{\sigma_Y}}_{\text{new slope}} Z_X + \frac{\epsilon}{\sigma_Y}. && \text{(definition of } Z_X\text{)}
\end{aligned}$$

$$\rho = \beta_1 \frac{\sigma_{X_1}}{\sigma_Y} = \frac{\operatorname{cov}(X_1, Y)}{\sigma_{X_1}\sigma_Y} \qquad = \text{the slope of } Z_Y \text{ on } Z_X.$$

2.7 Variance explained: $\rho^2 = 1 - \operatorname{var}(Y \mid X)/\operatorname{var}(Y)$

Method

Goal: ρ^2 as a fraction of variance.
Start from: the variance equation $\sigma_Y^2 = \beta_1^2 \sigma_{X_1}^2 + \sigma_\epsilon^2$ (Section 2.4).
Moves: divide through by σ_Y^2; recognize $\beta_1^2 \sigma_{X_1}^2/\sigma_Y^2 = \rho^2$ (square Section 2.6); replace σ_ϵ^2 by $\operatorname{var}(Y \mid X)$ (Section 2.5).

$$\begin{aligned}
\sigma_Y^2 &= \beta_1^2 \sigma_{X_1}^2 + \sigma_\epsilon^2 && \text{(variance equation)} \\
1 &= \frac{\beta_1^2 \sigma_{X_1}^2}{\sigma_Y^2} + \frac{\sigma_\epsilon^2}{\sigma_Y^2} && \text{(divide by } \sigma_Y^2\text{)} \\
1 &= \rho^2 + \frac{\sigma_\epsilon^2}{\sigma_Y^2} && (\rho^2 = (\beta_1 \sigma_{X_1}/\sigma_Y)^2) \\
\rho^2 &= 1 - \frac{\operatorname{var}(Y \mid X)}{\operatorname{var}(Y)} && (\sigma_\epsilon^2 = \operatorname{var}(Y \mid X),\ \sigma_Y^2 = \operatorname{var} Y)
\end{aligned}$$

$$\rho^2 = 1 - \frac{\text{var}(Y \mid X)}{\text{var}(Y)} = \frac{\beta_1^2 \sigma_{X_1}^2}{\sigma_Y^2}.$$

Bonus: why $-1 \le \rho \le 1$. Both pieces ρ^2 and $\sigma_\epsilon^2/\sigma_Y^2$ are ≥ 0 and add to 1, so $\rho^2 \le 1$.

3 Two (or more) predictors

Now the model is

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon, \qquad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2), \qquad \epsilon \text{ independent of } X_1, X_2.$$

β_1 is the effect of X_1 *with* X_2 *held fixed*. We write β_1' for the slope when Y is regressed on X_1 alone.

3.1 Conditional variance and ρ^2 with two predictors; adding a predictor never lowers ρ^2

Method

Goal: (a) $\text{var}(Y \mid X_1, X_2)$; (b) show ρ^2 with both predictors $\geq \rho^2$ with X_1 alone.
Start from: the two-predictor model.
Moves: M5 condition on what is known; M4 the remaining random pieces are independent, so variances add; compare.

(a) Given both X_1 and X_2, the whole $\beta_0 + \beta_1 X_1 + \beta_2 X_2$ is a constant, so exactly as in Section 2.5,

$$\text{var}(Y \mid X_1, X_2) = \text{var}(\epsilon) = \sigma_\epsilon^2, \qquad \text{so} \qquad \rho^2 = 1 - \frac{\text{var}(Y \mid X_1, X_2)}{\text{var}\, Y} = 1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2}.$$

(Taking var of the model, with M4's general rule since X_1, X_2 may be correlated: $\sigma_Y^2 = \beta_1^2 \sigma_{X_1}^2 + \beta_2^2 \sigma_{X_2}^2 + 2\beta_1\beta_2 \,\text{cov}(X_1, X_2) + \sigma_\epsilon^2$.)

(b) Given only X_1, the term $\beta_0 + \beta_1 X_1$ is a constant but X_2 and ϵ are still random:

$$\begin{aligned} \text{var}(Y \mid X_1) &= \text{var}(\beta_2 X_2 + \epsilon \mid X_1) && \text{(drop the known constant)} \\ &= \beta_2^2 \,\text{var}(X_2 \mid X_1) + \sigma_\epsilon^2 && (\epsilon \text{ independent of } X_1, X_2\text{: variances add}) \\ &\geq \sigma_\epsilon^2 = \text{var}(Y \mid X_1, X_2). && \text{(a variance is } \geq 0) \end{aligned}$$

Since $\rho^2 = 1 - \text{var}(Y \mid \cdots)/\,\text{var}\, Y$, a smaller conditional variance means a bigger ρ^2:

$$\rho^2_{\text{with } X_1, X_2} \;\geq\; \rho^2_{\text{with } X_1 \text{ only}}.$$

Adding a predictor never decreases ρ^2 (and the same argument with sample quantities shows R^2 never decreases).

3.2 Omitted-variable bias: $\beta_1' = \beta_1 + \beta_2 \beta_{1,2}$

Method

Goal: the slope β_1' you get if you regress Y on X_1 alone, when the truth has two predictors.
Start from: the one-predictor formula $\beta_1' = \text{cov}(X_1, Y)/\,\text{var}(X_1)$ (Section 2.1), which holds for *any* Y.
Moves: M1 substitute the *two*-predictor model for Y; M3 split the covariance; divide by $\text{var}\, X_1$; name the leftover ratio $\beta_{1,2}$.

$$\begin{aligned}\operatorname{cov}(X_1, Y) &= \operatorname{cov}(X_1,\ \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \epsilon) && \text{(substitute the two-predictor model)}\\ &= \beta_1 \operatorname{cov}(X_1, X_1) + \beta_2 \operatorname{cov}(X_1, X_2) + \operatorname{cov}(X_1, \epsilon) && \text{(bilinearity; } \beta_0 \text{ drops out)}\\ &= \beta_1 \operatorname{var}(X_1) + \beta_2 \operatorname{cov}(X_1, X_2). && \text{(}\operatorname{cov}(X_1, \epsilon) = 0\text{)}\end{aligned}$$

Divide by $\operatorname{var}(X_1)$:

$$\beta_1' = \frac{\operatorname{cov}(X_1, Y)}{\operatorname{var}(X_1)} = \beta_1 + \beta_2 \frac{\operatorname{cov}(X_1, X_2)}{\operatorname{var}(X_1)}.$$

The last fraction has the same form as Section 2.1 with X_2 in the role of Y: it is the slope when X_2 is regressed on X_1.

Omitted-variable bias

$$\beta_1' = \beta_1 + \beta_2\,\beta_{1,2}, \qquad \beta_{1,2} = \frac{\operatorname{cov}(X_1, X_2)}{\operatorname{var}(X_1)} \quad \text{(slope of } X_2 \text{ on } X_1\text{)}.$$

The bias $\beta_2\beta_{1,2}$ is 0 only if $\beta_2 = 0$ (no effect) or $\beta_{1,2} = 0$ (uncorrelated, as in a randomized trial).

(The notes assume mean-zero predictors and no intercept "for simplicity". As the steps above show, that is not needed: constants always drop out of a covariance.)

Method 2: conditional means (when X_1 is binary). This is how Lecture 12 explained the test-score example. Condition on X_1 M5 and take $\mathbb{E}$ M2:

$$\mathbb{E}[Y \mid X_1] = \beta_0 + \beta_1 X_1 + \beta_2 \mathbb{E}[X_2 \mid X_1] + \underbrace{\mathbb{E}[\epsilon \mid X_1]}_{0}.$$

Subtract the $X_1 = 0$ version from the $X_1 = 1$ version (as in Section 2.3):

$$\underbrace{\mathbb{E}[Y \mid X_1{=}1] - \mathbb{E}[Y \mid X_1{=}0]}_{\beta_1'} = \beta_1 + \beta_2 \underbrace{\left(\mathbb{E}[X_2 \mid X_1{=}1] - \mathbb{E}[X_2 \mid X_1{=}0]\right)}_{\beta_{1,2}}.$$

Same formula. Check with the numbers: $11.8 \approx 5.95 + 0.56 \times 10.3$.

Trick from the crime example: two OVB slopes give the predictor correlation

$\beta_{1,2}\,\beta_{2,1} = \dfrac{\operatorname{cov}(X_1, X_2)}{\operatorname{var} X_1} \cdot \dfrac{\operatorname{cov}(X_1, X_2)}{\operatorname{var} X_2} = \left(\dfrac{\operatorname{cov}(X_1, X_2)}{\sigma_{X_1}\sigma_{X_2}}\right)^2 = \rho_{1,2}^2$. So if you solve OVB both ways for the two slopes, $\rho_{1,2} = \pm\sqrt{\beta_{1,2}\beta_{2,1}}$, with the sign of the slopes.

3.3 (extra) Omitted-variable bias with K predictors

Method

Goal: β_1' when $X_2, \ldots, X_K$ are all left out.
Start from: $\beta_1' = \operatorname{cov}(X_1, Y)/\operatorname{var}(X_1)$.
Moves: identical to Section 3.2: M1 M3, now with a sum of left-out terms.

$$\begin{aligned}\operatorname{cov}(X_1, Y) &= \operatorname{cov}\Big(X_1,\ \beta_0 + \beta_1 X_1 + \sum_{j=2}^{K} \beta_j X_j + \epsilon\Big) && \text{(substitute the model)}\\ &= \beta_1 \operatorname{var}(X_1) + \sum_{j=2}^{K} \beta_j \operatorname{cov}(X_1, X_j) + 0 && \text{(bilinearity, one term at a time)}\end{aligned}$$

$$\beta_1' = \beta_1 + \sum_{j=2}^{K} \beta_j\, \beta_{1,j}, \qquad \beta_{1,j} = \frac{\operatorname{cov}(X_1, X_j)}{\operatorname{var}(X_1)}.$$

Picture: every path from X_1 to Y adds up: the direct arrow β_1, plus each detour $X_1 \to X_j \to Y$ worth $\beta_{1,j}\beta_j$.

3.4 (extra) The covariance system: solving for β_1 and β_2

Method

Goal: the two-predictor coefficients in terms of covariances.
Start from: the two-predictor model.
Moves: M3 *twice*: take $\operatorname{cov}(X_1, \cdot)$ and $\operatorname{cov}(X_2, \cdot)$ of the model. That gives two equations in two unknowns; eliminate β_2.

Write $c = \operatorname{cov}(X_1, X_2)$. The first line was done in Section 3.2; the second is the same move with X_2:

$$\operatorname{cov}(X_1, Y) = \beta_1 \sigma^2_{X_1} + \beta_2\, c \tag{i}$$
$$\operatorname{cov}(X_2, Y) = \beta_1\, c + \beta_2 \sigma^2_{X_2} \tag{ii}$$

Eliminate β_2: multiply (i) by $\sigma^2_{X_2}$ and (ii) by c, then subtract:

$$\sigma^2_{X_2} \operatorname{cov}(X_1, Y) - c \operatorname{cov}(X_2, Y) = \beta_1 \big(\sigma^2_{X_1}\sigma^2_{X_2} - c^2\big).$$

$$\beta_1 = \frac{\operatorname{cov}(X_1, Y)\, \sigma^2_{X_2} - \operatorname{cov}(X_2, Y)\, \operatorname{cov}(X_1, X_2)}{\sigma^2_{X_1}\sigma^2_{X_2} - \operatorname{cov}(X_1, X_2)^2} \overset{\text{standardized}}{=} \frac{\rho_1 - \rho_{1,2}\,\rho_2}{1 - \rho^2_{1,2}},$$

where $\rho_k = \operatorname{corr}(X_k, Y)$ and $\rho_{1,2} = \operatorname{corr}(X_1, X_2)$.

For the standardized version, put all of X_1, X_2, Y in SD units M6: then every $\sigma = 1$, $\operatorname{cov}(X_k, Y) = \rho_k$, and $c = \rho_{1,2}$. **Checks:** (1) dividing (i) by $\sigma^2_{X_1}$ is exactly OVB; (2) if $\rho_{1,2} = 0$ then $\beta_1 = \rho_1$, the single-predictor slope.

Matrix form (as in the notes). (i)–(ii) say $\big(\operatorname{cov}(X_1, Y), \operatorname{cov}(X_2, Y)\big)^T = \Sigma\,(\beta_1, \beta_2)^T$ with $\Sigma = \begin{bmatrix} \sigma^2_{X_1} & c \\ c & \sigma^2_{X_2} \end{bmatrix}$, so $(\beta_1, \beta_2)^T = \Sigma^{-1}\big(\operatorname{cov}(X_1, Y), \operatorname{cov}(X_2, Y)\big)^T$ using $\begin{bmatrix} a & b \\ b & d \end{bmatrix}^{-1} = \frac{1}{ad - b^2}\begin{bmatrix} d & -b \\ -b & a \end{bmatrix}$. It gives the same β_1.

3.5 (extra) Correlated predictors inflate the standard error

Method

Goal: $\text{se}(\hat{\beta}_1)$ when X_2 is also in the model.
Start from: the result stated in the notes (Unit 4.3–4.4): the variances and covariance of $(\hat{\beta}_1, \hat{\beta}_2)$ are $\frac{\sigma_\epsilon^2}{N}\Sigma^{-1}$.
Moves: invert Σ (2×2 rule); read off the diagonal for variances and the off-diagonal for the covariance; simplify with $c = \rho_{1,2}\sigma_{X_1}\sigma_{X_2}$.

Since $c = \rho_{1,2}\sigma_{X_1}\sigma_{X_2}$, the determinant is $\sigma_{X_1}^2\sigma_{X_2}^2 - c^2 = \sigma_{X_1}^2\sigma_{X_2}^2(1-\rho_{1,2}^2)$. The top-left entry of Σ^{-1} is

$$\frac{\sigma_{X_2}^2}{\sigma_{X_1}^2\sigma_{X_2}^2(1-\rho_{1,2}^2)} = \frac{1}{\sigma_{X_1}^2(1-\rho_{1,2}^2)} \quad \Longrightarrow \quad \text{var}(\hat{\beta}_1) = \frac{\sigma_\epsilon^2}{N\sigma_{X_1}^2(1-\rho_{1,2}^2)}.$$

$$\text{se}(\hat{\beta}_1) = \frac{\sigma_\epsilon}{\sigma_{X_1}\sqrt{N(1-\rho_{1,2}^2)}} = \underbrace{\frac{\sigma_\epsilon}{\sigma_{X_1}\sqrt{N}}}_{\text{one predictor}} \cdot \frac{1}{\sqrt{1-\rho_{1,2}^2}}, \qquad \text{corr}(\hat{\beta}_1, \hat{\beta}_2) = -\rho_{1,2}.$$

Correlation of the estimates (standardized predictors): $\Sigma^{-1} = \frac{1}{1-\rho_{1,2}^2}\begin{bmatrix} 1 & -\rho_{1,2} \\ -\rho_{1,2} & 1 \end{bmatrix}$, so corr $= \dfrac{\text{cov}}{\text{se}\cdot\text{se}} = \dfrac{-\rho_{1,2}/(1-\rho_{1,2}^2)}{1/(1-\rho_{1,2}^2)} = -\rho_{1,2}.$

Method 2 (intuition you can rebuild from the sheet). Only the part of X_1 that X_2 cannot predict helps separate β_1 from β_2. Apply the variance-explained formula (Section 2.7) to X_1 regressed on X_2: the leftover variance is $\text{var}(X_1 \mid X_2) = (1-\rho_{1,2}^2)\sigma_{X_1}^2$. Put that leftover SD in place of σ_{X_1} in $\text{se} = \sigma_\epsilon/(\sigma_{X_1}\sqrt{N})$ and you get the boxed formula.

4 Estimators and their sample distributions

4.1 Definitions (nothing to derive)

$Y_1, \dots, Y_N$ iid; an estimator $\hat{\theta} = \hat{\theta}(Y_1, \dots, Y_N)$ is any function of the data. Over replicate datasets it varies; its distribution over replicates is the **sample distribution** and its standard deviation is $\text{se}(\hat{\theta}) = \sqrt{\text{var}(\hat{\theta})}$. $\text{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$. These are definitions: learn them, do not derive them.

4.2 The sample mean: $\overline{Y} \approx \text{Normal}(\mu, \sigma^2/N)$, $\text{se}(\overline{Y}) = \sigma/\sqrt{N}$

Method

Goal: the sample distribution of $\overline{Y}$ when $\mathbb{E}Y_i = \mu$, $\text{var}\, Y_i = \sigma^2$.
Start from: $\overline{Y} = \frac{1}{N}\sum_i Y_i$.
Moves: the general recipe: M2 center, M4 spread (independence: variances add), M7 shape.

Center.

$$\mathbb{E}[\overline{Y}] = \mathbb{E}\Big[\frac{1}{N}\sum_i Y_i\Big] = \frac{1}{N}\sum_i \mathbb{E}[Y_i] = \frac{1}{N}\cdot N\mu = \mu.$$

Spread.

$$\begin{aligned} \text{var}(\overline{Y}) &= \text{var}\left(\frac{1}{N}\sum_i Y_i\right) = \frac{1}{N^2}\,\text{var}\left(\sum_i Y_i\right) && (\text{var}(aX) = a^2 \text{var}\, X \text{ with } a = 1/N) \\ &= \frac{1}{N^2}\sum_i \text{var}(Y_i) = \frac{1}{N^2}\cdot N\sigma^2 = \frac{\sigma^2}{N} && (\text{independent: variances add}) \end{aligned}$$

Shape. If the Y_i are Normal, $\sum_i Y_i$ is exactly Normal (a sum of independent Normals). Otherwise the CLT says $\overline{Y}$ is approximately Normal for large N.

$$\overline{Y} \approx \text{Normal}\Big(\mu,\ \frac{\sigma^2}{N}\Big), \qquad \text{se}(\overline{Y}) = \frac{\sigma}{\sqrt{N}}.$$

Unbiased ($\mathbb{E}\overline{Y} = \mu$) and consistent ($\text{se} \to 0$).

Bernoulli version. If $Y_i \sim \text{Bernoulli}(q)$ then $\mu = q$ and $\sigma^2 = q(1-q)$, so $\hat{q} = \overline{Y} \approx \text{Normal}\big(q,\ q(1-q)/N\big)$.

Common slips: the $1/N$ comes out of the variance *squared* ($1/N^2$); se has $\sqrt{N}$, the variance has N.

4.3 Bias–variance decomposition: $\text{MSE} = \text{se}^2 + \text{bias}^2$

Method

Goal: split $\text{MSE}_{\hat{\theta}} = \mathbb{E}[(\hat{\theta} - \theta)^2]$ into variance and bias.
Start from: either the variance formula $\text{var}(W) = \mathbb{E}[W^2] - (\mathbb{E}W)^2$ (method A, the notes' proof) or the square itself (method B).
Moves: A: apply the formula to $W = \hat{\theta} - \theta$. B: M10 add and subtract $\mathbb{E}[\hat{\theta}]$, expand, show the cross term is 0.

Method A. Let $W = \hat{\theta} - \theta$. Since θ is a fixed number, $\mathrm{var}(W) = \mathrm{var}(\hat{\theta})$ and $\mathbb{E}W = \mathbb{E}\hat{\theta} - \theta$.

$$\begin{aligned} \mathrm{var}(W) &= \mathbb{E}[W^2] - (\mathbb{E}W)^2 && \text{(variance formula)} \\ \mathrm{var}(\hat{\theta}) &= \mathbb{E}\big[(\hat{\theta}-\theta)^2\big] - \big(\mathbb{E}[\hat{\theta}] - \theta\big)^2 && \text{(shifting by } \theta \text{ does not change variance)} \\ \mathrm{se}(\hat{\theta})^2 &= \mathrm{MSE}_{\hat{\theta}} - \mathrm{bias}(\hat{\theta})^2. && \text{(definitions)} \end{aligned}$$

Method B. Write $m = \mathbb{E}[\hat{\theta}]$ (a constant).

$$\begin{aligned} \mathbb{E}\big[(\hat{\theta}-\theta)^2\big] &= \mathbb{E}\big[\big((\hat{\theta}-m) + (m-\theta)\big)^2\big] && \text{(add and subtract } m\text{)} \\ &= \mathbb{E}\big[(\hat{\theta}-m)^2\big] + 2(m-\theta)\,\mathbb{E}[\hat{\theta}-m] + (m-\theta)^2 && \text{(expand; } m-\theta \text{ is a constant)} \\ &= \mathrm{var}(\hat{\theta}) + 0 + (m-\theta)^2 && (\mathbb{E}[\hat{\theta}-m] = m - m = 0) \end{aligned}$$

$$\mathrm{MSE}_{\hat{\theta}} = \mathrm{se}(\hat{\theta})^2 + \mathrm{bias}(\hat{\theta})^2.$$

4.4 (extra) Bessel's correction: why divide by $N-1$

Method

Goal: $\mathbb{E}\big[\sum_i (Y_i - \overline{Y})^2\big]$.
Start from: the sum of squares $\sum_i (Y_i - \overline{Y})^2$.
Moves: M10 add and subtract μ inside the square; expand; use $\sum_i (Y_i - \mu) = N(\overline{Y} - \mu)$; take $\mathbb{E}$ using $\mathbb{E}[(Y_i-\mu)^2] = \sigma^2$ and $\mathbb{E}[(\overline{Y}-\mu)^2] = \mathrm{var}\,\overline{Y} = \sigma^2/N$.

$$\begin{aligned} \sum_i (Y_i - \overline{Y})^2 &= \sum_i \big[(Y_i - \mu) - (\overline{Y} - \mu)\big]^2 && \text{(add and subtract } \mu\text{)} \\ &= \sum_i (Y_i-\mu)^2 - 2(\overline{Y}-\mu)\sum_i (Y_i - \mu) + N(\overline{Y}-\mu)^2 && \text{(expand; } \overline{Y}-\mu \text{ same for all } i\text{)} \\ &= \sum_i (Y_i-\mu)^2 - 2N(\overline{Y}-\mu)^2 + N(\overline{Y}-\mu)^2 && \big(\textstyle\sum_i (Y_i-\mu) = N(\overline{Y}-\mu)\big) \\ &= \sum_i (Y_i-\mu)^2 - N(\overline{Y}-\mu)^2. \end{aligned}$$

Take expectations: $\mathbb{E}\big[\sum_i (Y_i - \overline{Y})^2\big] = N\sigma^2 - N \cdot \frac{\sigma^2}{N} = (N-1)\sigma^2$.

$$\hat{\sigma}_0^2 = \tfrac{1}{N}\textstyle\sum_i (Y_i - \overline{Y})^2 \text{ has } \mathbb{E}[\hat{\sigma}_0^2] = \tfrac{N-1}{N}\sigma^2; \qquad \hat{\sigma}^2 = \tfrac{1}{N-1}\textstyle\sum_i (Y_i - \overline{Y})^2 \text{ is unbiased.}$$

The same idea explains $\hat{\sigma}_\epsilon^2 = \mathrm{SSR}/(N-2)$: two parameters were fitted from the data, so divide by $N-2$.

5 Confidence intervals and hypothesis tests

5.1 The 95% confidence interval: $\hat{\theta} \pm 1.96\ \mathrm{se}(\hat{\theta})$

Method

Goal: an interval that contains θ in 95% of replicates.
Start from: the sample distribution $\hat{\theta} \approx \mathrm{Normal}(\theta, \mathrm{se}^2)$. This needs $\hat{\theta}$ **unbiased**, so the center is θ.
Moves: M6 standardize; M9 95% of a standard Normal lies in $(-1.96, 1.96)$; rearrange the inequality to isolate θ.

$$\frac{\hat{\theta} - \theta}{\mathrm{se}(\hat{\theta})} \approx \mathrm{Normal}(0,1) \qquad \text{(standardize; uses } \mathbb{E}\hat{\theta} = \theta\text{)}$$

$$\mathbb{P}\Big(-1.96 < \frac{\hat{\theta} - \theta}{\mathrm{se}(\hat{\theta})} < 1.96\Big) \approx 0.95 \qquad \text{(}\mathbb{P}(|Z| < 1.96) = 0.95\text{)}$$

$$\mathbb{P}\big(-1.96\ \mathrm{se} < \hat{\theta} - \theta < 1.96\ \mathrm{se}\big) \approx 0.95 \qquad \text{(multiply through by se} > 0\text{)}$$

$$\mathbb{P}\big(\hat{\theta} - 1.96\ \mathrm{se} < \theta < \hat{\theta} + 1.96\ \mathrm{se}\big) \approx 0.95 \qquad \text{(rearrange both inequalities for } \theta\text{)}$$

$$95\%\ \mathrm{CI} = \big[\hat{\theta} - 1.96\ \mathrm{se}(\hat{\theta}),\ \hat{\theta} + 1.96\ \mathrm{se}(\hat{\theta})\big] \qquad \text{(when } \hat{\theta} \text{ is unbiased).}$$

The random thing is the interval, not θ: over replicates, 95% of such intervals contain θ.

Why "unbiased" matters. If $\mathbb{E}\hat{\theta} \neq \theta$, the bell curve is centered in the wrong place and the step "standardize" fails: the interval covers θ less than 95% of the time.

(extra) How much data for a target width. If $\mathrm{se} = \sigma/\sqrt{N}$ and you want half-width at most h: $1.96\,\sigma/\sqrt{N} \le h \iff \sqrt{N} \ge 1.96\,\sigma/h \iff N \ge (1.96\,\sigma/h)^2$.

5.2 The test statistic and p-value

Method

Goal: how surprising $\hat{\beta}_1$ would be if $H_0 : \beta_1 = 0$ were true.
Start from: the sample distribution $\hat{\beta}_1 \approx \mathrm{Normal}(\beta_1, \mathrm{se}^2)$ (Section 6.4).
Moves: set $\beta_1 = 0$ (the null); M6 standardize by se; M9 two tails of a standard Normal.

Under H_0, a replicate estimate $\hat{\beta}_1^* \approx \mathrm{Normal}(0, \mathrm{se}(\hat{\beta}_1)^2)$. Dividing by se M6:

$$\hat{T}^* = \frac{\hat{\beta}_1^*}{\mathrm{se}(\hat{\beta}_1)} \approx \mathrm{Normal}(0,1), \qquad \text{and for our data} \qquad \hat{T} = \frac{\hat{\beta}_1}{\mathrm{se}(\hat{\beta}_1)}.$$

The p-value is the chance a null replicate is at least as extreme as ours. Split it into two tails and use symmetry of $\mathrm{Normal}(0,1)$:

$$\begin{aligned} p_v = \mathbb{P}\big(|\hat{T}^*| > |\hat{T}|\big) &= \mathbb{P}\big(\hat{T}^* > |\hat{T}|\big) + \mathbb{P}\big(\hat{T}^* < -|\hat{T}|\big) && \text{(two tails)} \\ &= 2\big(1 - \Phi(|\hat{T}|)\big) && \text{(both tails have the same area)} \end{aligned}$$

$$H_0 : \beta_1 = 0, \qquad \hat{T} = \frac{\hat{\beta}_1}{\mathrm{se}(\hat{\beta}_1)}, \qquad p_v = \mathbb{P}\big(|\hat{T}^*| > |\hat{T}|\big) = 2\big(1 - \Phi(|\hat{T}|)\big).$$

By hand: near 0 the standard Normal density is about 0.4, so $\mathbb{P}(|Z| < a) \approx 0.8a$ for small a; then $p_v \approx 1 - 0.8|\hat{T}|$.

5.3 $p_v < 0.05 \iff 0 \notin$ 95% CI

Method

Goal: show the test and the CI always agree.
Start from: $\mathbb{P}(|Z| > 1.96) = 0.05$, and the larger $|\hat{T}|$ is, the smaller p_v is.
Moves: a chain of equivalent statements: $p_v \to |\hat{T}| \to |\hat{\beta}_1| \to$ the CI endpoints.

$$\begin{aligned}
p_v < 0.05 &\iff |\hat{T}| > 1.96 && (p_v = 0.05 \text{ at } |\hat{T}| = 1.96;\ p_v \text{ shrinks as } |\hat{T}| \text{ grows}) \\
&\iff |\hat{\beta}_1| > 1.96\ \mathrm{se}(\hat{\beta}_1) && (\text{multiply by se} > 0) \\
&\iff \hat{\beta}_1 - 1.96\ \mathrm{se} > 0 \ \text{ or } \ \hat{\beta}_1 + 1.96\ \mathrm{se} < 0 && (\hat{\beta}_1 > 1.96\ \mathrm{se} \text{ or } \hat{\beta}_1 < -1.96\ \mathrm{se}) \\
&\iff \text{CI entirely above 0 or entirely below 0} && (\text{these are the CI endpoints}) \\
&\iff 0 \notin 95\%\ \text{CI}.
\end{aligned}$$

6 Least squares and the sample distribution of $\hat{\beta}_1$

Data $(X_1, Y_1), \ldots, (X_N, Y_N)$ from the one-predictor model. (Here X_i means the i-th data point of the single predictor.)

6.1 The estimators $\hat{\beta}_1$ and $\hat{\beta}_0$

Method

Goal: estimates of β_1, β_0 from data.
Start from: the population formulas $\beta_1 = \mathrm{cov}(X,Y)/\sigma_X^2$ and $\beta_0 = \mathbb{E}Y - \beta_1 \mathbb{E}X$ (Sections 2.1–2.2).
Moves: M8 plug in: replace every $\mathbb{E}[\cdot]$ by $\frac{1}{N}\sum_i$ and $\mathbb{E}X, \mathbb{E}Y$ by $\overline{X}, \overline{Y}$. The $\frac{1}{N}$'s cancel.

Write $\mathrm{cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$ and $\sigma_X^2 = \mathbb{E}[(X - \mathbb{E}X)^2]$, then plug in:

$$\hat{\beta}_1 = \frac{\frac{1}{N}\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\frac{1}{N}\sum_i (X_i - \overline{X})^2} = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}, \qquad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}.$$

Method 2 (calculus; the "least squares" name). Minimize $f(b_0, b_1) = \sum_i (Y_i - b_0 - b_1 X_i)^2$: set the derivative in each of b_0, b_1 to 0.

$$\begin{aligned} &\text{in } b_0: \quad \textstyle\sum_i -2(Y_i - b_0 - b_1 X_i) = 0 \implies N\overline{Y} - Nb_0 - b_1 N\overline{X} = 0 \implies b_0 = \overline{Y} - b_1 \overline{X}, \\ &\text{in } b_1: \quad \textstyle\sum_i -2X_i(Y_i - b_0 - b_1 X_i) = 0. \end{aligned}$$

Substitute b_0 into the second equation: $\sum_i X_i\big[(Y_i - \overline{Y}) - b_1(X_i - \overline{X})\big] = 0$, so $b_1 = \frac{\sum_i X_i(Y_i - \overline{Y})}{\sum_i X_i(X_i - \overline{X})}$. Then the sum trick M11: subtracting $\overline{X}\sum_i (Y_i - \overline{Y}) = 0$ on top and $\overline{X}\sum_i (X_i - \overline{X}) = 0$ on the bottom turns X_i into $(X_i - \overline{X})$, giving the same $\hat{\beta}_1$.

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \overline{X})(Y_i - \overline{Y})}{\sum_i (X_i - \overline{X})^2}, \qquad \hat{\beta}_0 = \overline{Y} - \hat{\beta}_1 \overline{X}.$$

6.2 (extra) $\hat{\sigma}_\epsilon^2$ and R^2

Method

Goal: estimates of σ_ϵ^2 and ρ^2.
Start from: $\sigma_\epsilon^2 = \mathrm{var}(Y \mid X)$ and $\rho^2 = 1 - \mathrm{var}(Y \mid X)/\mathrm{var}(Y)$.
Moves: M8 plug in: the spread around the line is estimated by the residuals; the total spread by deviations from $\overline{Y}$.

Residuals $r_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$, $\mathrm{SSR} = \sum_i r_i^2$, $\mathrm{SST} = \sum_i (Y_i - \overline{Y})^2$. Then $\mathrm{var}(Y \mid X) \approx \mathrm{SSR}/N$ and $\mathrm{var}(Y) \approx \mathrm{SST}/N$, and the N's cancel in the ratio:

$$\hat{\sigma}_\epsilon^2 = \frac{\mathrm{SSR}}{N-2} \left(\approx \frac{\mathrm{SSR}}{N} \right), \qquad R^2 = 1 - \frac{\mathrm{SSR}}{\mathrm{SST}} \approx \rho^2.$$

(The $N - 2$ instead of N is Bessel's correction again, Section 4.4.)

6.3 $\hat{\beta}_1$ is unbiased: $\mathbb{E}[\hat{\beta}_1] = \beta_1$

Method

Goal: $\mathbb{E}[\hat{\beta}_1 \mid X_1, \ldots, X_N]$.
Start from: the formula for $\hat{\beta}_1$, treating the X's as fixed numbers M5.
Moves: M2 linearity moves $\mathbb{E}$ onto the Y's; M1 $\mathbb{E}[Y_i \mid X_i] = \beta_0 + \beta_1 X_i$; cancel.

Given the X's, the weights $(X_i - \overline{X})$ and the denominator $S_{xx} = \sum_i (X_i - \overline{X})^2$ are constants.

$$\mathbb{E}[\hat{\beta}_1 \mid X] = \frac{\sum_i (X_i - \overline{X})\, \mathbb{E}[Y_i - \overline{Y} \mid X]}{S_{xx}} \qquad \text{(linearity; constants come out)}$$

$$\mathbb{E}[Y_i - \overline{Y} \mid X] = (\beta_0 + \beta_1 X_i) - (\beta_0 + \beta_1 \overline{X}) = \beta_1 (X_i - \overline{X}) \qquad \text{(model, averaged over } i \text{ for } \overline{Y})$$

$$\mathbb{E}[\hat{\beta}_1 \mid X] = \frac{\sum_i (X_i - \overline{X})\, \beta_1 (X_i - \overline{X})}{S_{xx}} = \beta_1 \frac{S_{xx}}{S_{xx}} = \beta_1. \qquad \text{(substitute)}$$

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 \qquad \text{(conditional on the } X\text{'s)}.$$

The key ingredient is that the noise is independent of *all* the predictors (this fails for AR models: the Hurwicz bias).

6.4 The spread: $\mathrm{se}(\hat{\beta}_1) \approx \sigma_\epsilon / (\sigma_X \sqrt{N})$

Method

Goal: $\mathrm{var}(\hat{\beta}_1 \mid X)$, then the full sample distribution.
Start from: $\hat{\beta}_1$ written as a weighted sum of the Y_i.
Moves: M11 drop $\overline{Y}$ from the numerator; M4 independent Y_i's, each with $\mathrm{var}(Y_i \mid X_i) = \sigma_\epsilon^2$: variances add with weights squared; replace S_{xx} by $N\sigma_X^2$; shape from "a weighted sum of Normals is Normal".

Step 1: weighted sum. $\sum_i (X_i - \overline{X})\overline{Y} = \overline{Y} \sum_i (X_i - \overline{X}) = 0$ M11, so

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \overline{X}) Y_i}{S_{xx}} = \sum_i w_i Y_i, \qquad w_i = \frac{X_i - \overline{X}}{S_{xx}}.$$

Step 2: variance.

$$\mathrm{var}(\hat{\beta}_1 \mid X) = \sum_i w_i^2 \,\mathrm{var}(Y_i \mid X_i) \qquad \text{(independent } Y_i\text{; } \mathrm{var}(w_i Y_i) = w_i^2 \,\mathrm{var}\, Y_i)$$

$$= \frac{\sum_i (X_i - \overline{X})^2}{S_{xx}^2} \sigma_\epsilon^2 = \frac{S_{xx}}{S_{xx}^2} \sigma_\epsilon^2 = \frac{\sigma_\epsilon^2}{S_{xx}} \qquad (\mathrm{var}(Y_i \mid X_i) = \sigma_\epsilon^2)$$

$$\approx \frac{\sigma_\epsilon^2}{N\sigma_X^2} \qquad (S_{xx}/N \text{ is the sample variance of } X, \approx \sigma_X^2)$$

Step 3: shape. Given the X's, each Y_i is Normal, so the weighted sum $\hat{\beta}_1$ is Normal.

$$\mathbb{E}[\hat{\beta}_1] = \beta_1, \qquad \text{se}(\hat{\beta}_1) \approx \frac{\sigma_\epsilon}{\sigma_X\sqrt{N}}, \qquad \hat{\beta}_1 \approx \text{Normal}\Big(\beta_1,\ \frac{\sigma_\epsilon^2}{N\sigma_X^2}\Big).$$

More precise with less noise, more spread-out X's, or more data.

Compare with $\overline{Y}$. The recipe is identical: $\overline{Y}$ has weights $1/N$ (giving σ^2/N); $\hat{\beta}_1$ has weights w_i (giving σ_ϵ^2/S_{xx}).

6.5 (extra) Check: a binary predictor gives a difference of means

Method

Goal: the same se by a second route when $X \sim \text{Bernoulli}(p)$.
Start from: $\hat{\beta}_1 = \overline{Y}_{X=1} - \overline{Y}_{X=0}$ (Section 2.3).
Moves: each group mean is $\overline{Y}$ for that group M7; the groups are independent, so variances add M4 (even though we subtract).

With Np points in group 1 and $N(1-p)$ in group 0,

$$\begin{aligned}\text{var}(\hat{\beta}_1) &= \frac{\sigma_\epsilon^2}{Np} + \frac{\sigma_\epsilon^2}{N(1-p)} && (\text{var}(A-B) = \text{var}\,A + \text{var}\,B \text{ for independent } A, B)\\ &= \sigma_\epsilon^2\,\frac{(1-p)+p}{Np(1-p)} = \frac{\sigma_\epsilon^2}{N\,p(1-p)} = \frac{\sigma_\epsilon^2}{N\sigma_X^2} && (\sigma_X^2 = p(1-p) \text{ for a Bernoulli})\end{aligned}$$

This matches Section 6.4. With equal groups ($p = \frac{1}{2}$) it is $4\sigma_\epsilon^2/N$, the clinical-trial result.

6.6 (extra) $\hat{T}^2 = (N-2)\,R^2/(1-R^2)$

Method

Goal: connect the test statistic to R^2.
Start from: $r_i = (Y_i - \overline{Y}) - \hat{\beta}_1(X_i - \overline{X})$ (from $\hat{\beta}_0 = \overline{Y} - \hat{\beta}_1\overline{X}$), and $\text{se}(\hat{\beta}_1)^2 = \hat{\sigma}_\epsilon^2/S_{xx}$.
Moves: square and sum r_i to get $\text{SSR} = \text{SST} - \hat{\beta}_1^2 S_{xx}$; substitute into $\hat{T}^2$; divide top and bottom by SST.

Using $\sum_i (X_i - \overline{X})(Y_i - \overline{Y}) = \hat{\beta}_1 S_{xx}$ (the definition of $\hat{\beta}_1$):

$$\begin{aligned}\text{SSR} &= \text{SST} - 2\hat{\beta}_1 \sum_i (X_i - \overline{X})(Y_i - \overline{Y}) + \hat{\beta}_1^2 S_{xx} && (\text{expand } \textstyle\sum_i r_i^2)\\ &= \text{SST} - 2\hat{\beta}_1^2 S_{xx} + \hat{\beta}_1^2 S_{xx} = \text{SST} - \hat{\beta}_1^2 S_{xx}. && (\text{substitute})\end{aligned}$$

So $\hat{\beta}_1^2 S_{xx} = \text{SST} - \text{SSR}$. With $\text{se}^2 = \text{SSR}/((N-2)S_{xx})$:

$$\begin{aligned}\hat{T}^2 = \frac{\hat{\beta}_1^2}{\text{se}^2} &= \frac{(N-2)\hat{\beta}_1^2 S_{xx}}{\text{SSR}} = \frac{(N-2)(\text{SST} - \text{SSR})}{\text{SSR}} && (\text{substitute})\\ &= (N-2)\frac{1 - \text{SSR}/\text{SST}}{\text{SSR}/\text{SST}} = (N-2)\frac{R^2}{1-R^2}. && (\text{divide top and bottom by SST})\end{aligned}$$

Use: a tiny p_v with a small R^2 is possible (large N); a large p_v with R^2 near 1 is not.

7 Self-test: cover the right two columns

For each formula, say out loud what you start from and the key move, then check. If you can do every row, you can derive the whole sheet.

Result	**Start from**	**Key move(s)**
$\beta_1 = \mathrm{cov}(X_1, Y)/\sigma^2_{X_1}$	model	$\mathrm{cov}(X_1, \cdot)$ of both sides; β_0 and ϵ drop out
$\beta_0 = \mathbb{E}Y - \beta_1 \mathbb{E}X_1$	model	$\mathbb{E}$ of both sides; $\mathbb{E}\epsilon = 0$
binary X: β_1 = difference of group means	model given $X = 1$ and $X = 0$	condition, take $\mathbb{E}$, subtract
$\sigma^2_Y = \beta_1^2 \sigma^2_{X_1} + \sigma^2_\epsilon$	model	var of both sides; independent pieces add; coefficient squared
$\mathrm{var}(Y \mid X_1) = \sigma^2_\epsilon$	model	given X_1, $\beta_0 + \beta_1 X_1$ is a constant
$\rho = \beta_1 \sigma_{X_1}/\sigma_Y$	$\rho = \mathrm{cov}\,/(\sigma_X \sigma_Y)$, or standardized model	substitute $\mathrm{cov} = \beta_1 \sigma^2_X$; or standardize and read the slope
$\rho^2 = 1 - \mathrm{var}(Y \mid X)/\mathrm{var}\,Y$	variance equation	divide by σ^2_Y; first term is ρ^2
ρ^2 never decreases	$\mathrm{var}(Y \mid X_1)$ in the 2-predictor model	$= \beta_2^2 \,\mathrm{var}(X_2 \mid X_1) + \sigma^2_\epsilon \geq \sigma^2_\epsilon$
OVB $\beta_1' = \beta_1 + \beta_2 \beta_{1,2}$	$\beta_1' = \mathrm{cov}(X_1, Y)/\mathrm{var}\,X_1$	substitute the *2-predictor* model; bilinearity; divide
OVB, K predictors	same	same, one term per left-out X_j
$\beta_{1,2}\beta_{2,1} = \rho^2_{1,2}$	definitions of the slopes	multiply; $\mathrm{cov}^2\,/(\mathrm{var}\,\mathrm{var})$
$\beta_1 = \frac{\rho_1 - \rho_{1,2}\rho_2}{1-\rho^2_{1,2}}$	model	$\mathrm{cov}(X_1, \cdot)$ *and* $\mathrm{cov}(X_2, \cdot)$; 2 equations; eliminate β_2
se inflation $1/\sqrt{1-\rho^2_{1,2}}$	$\frac{\sigma^2_\epsilon}{N}\Sigma^{-1}$ (given)	2×2 inverse; determinant $= \sigma_1^2\sigma_2^2(1-\rho^2_{1,2})$
$\overline{Y} \approx \mathrm{Normal}(\mu, \sigma^2/N)$	$\overline{Y} = \frac{1}{N}\sum Y_i$	$\mathbb{E}$: linearity; var: $\frac{1}{N^2} \cdot N\sigma^2$; shape: CLT
$\mathrm{MSE} = \mathrm{se}^2 + \mathrm{bias}^2$	$\mathrm{var}\,W = \mathbb{E}W^2 - (\mathbb{E}W)^2$	$W = \hat{\theta} - \theta$; or add and subtract $\mathbb{E}\hat{\theta}$
Bessel $(N-1)$	$\sum (Y_i - \overline{Y})^2$	add and subtract μ; $= \sum (Y_i - \mu)^2 - N(\overline{Y} - \mu)^2$
95% CI $\hat{\theta} \pm 1.96$ se	$\hat{\theta} \approx \mathrm{Normal}(\theta, \mathrm{se}^2)$ (unbiased)	standardize; $\mathbb{P}(\lvert Z\rvert < 1.96) = .95$; solve for θ
$\hat{T}$, $p_v = 2(1 - \Phi(\lvert\hat{T}\rvert))$	$\hat{\beta}_1^* \approx \mathrm{Normal}(0, \mathrm{se}^2)$ under H_0	divide by se; two equal tails
$p_v < .05 \iff 0 \notin \mathrm{CI}$	$\mathbb{P}(\lvert Z\rvert > 1.96) = .05$	$\lvert\hat{T}\rvert > 1.96 \iff \lvert\hat{\beta}_1\rvert > 1.96$ se $\iff$ CI misses 0
$\hat{\beta}_1, \hat{\beta}_0$	$\beta_1 = \mathrm{cov}\,/\sigma^2_X$, $\beta_0 = \mathbb{E}Y - \beta_1 \mathbb{E}X$	plug in sample averages; or minimize SSR
$\hat{\sigma}^2_\epsilon$, R^2	$\sigma^2_\epsilon = \mathrm{var}(Y \mid X)$, ρ^2	plug in SSR/N and SST/N
$\mathbb{E}\hat{\beta}_1 = \beta_1$	$\hat{\beta}_1$ formula, X's fixed	linearity; $\mathbb{E}[Y_i - \overline{Y} \mid X] = \beta_1(X_i - \overline{X})$
$\mathrm{se}(\hat{\beta}_1) \approx \sigma_\epsilon/(\sigma_X\sqrt{N})$	$\hat{\beta}_1 = \sum w_i Y_i$	sum trick drops $\overline{Y}$; $\mathrm{var} = \sum w_i^2 \sigma^2_\epsilon = \sigma^2_\epsilon/S_{xx}$
binary-X check	difference of group means	independent groups: variances add; $\sigma^2_X = p(1-p)$
$\hat{T}^2 = (N-2)\frac{R^2}{1-R^2}$	$r_i = (Y_i - \overline{Y}) - \hat{\beta}_1(X_i - \overline{X})$	$\mathrm{SSR} = \mathrm{SST} - \hat{\beta}_1^2 S_{xx}$; divide by SST